# Calibrating a commercial LLM judge for Filipino and Taglish evaluation: one-directional trust, and why we will not distil it

**PinoAI research note · JUDGE-001 · H-OPS-JUDGE**

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## Abstract

PH-EVAL, our native-authored Filipino and Taglish evaluation, has many free-generation items that exact-match checks cannot score. We score them with a commercial closed-API LLM judge (used zero-shot) that answers narrow, bounded questions, and a verdict counts only when the judge is confident. Everything else goes to a native-speaker reviewer. This note reports what two blind calibrations against that reviewer taught us.

- **Calibration #1 (n = 20 confident verdicts).** The judge agreed with the reviewer on 14 of 20 (70%, Cohen's κ = 0.37). Its errors were one-sided: 5 false passes against 1 false fail. It agreed on all 9 Taglish items.
- **Policy consequence.** The judge's FAIL is accepted. A confident PASS on privacy, hallucination and instruction following is escalated to a human.
- **Calibration #2 (72 blind rows).** The judge's Taglish naturalness probability ranks answers usefully (AUC 0.767 against the reviewer's pass/fail). A threshold of 0.7 gave 8 confident passes with 0 errors, but its fail side was wrong 2 times in 7.
- **A retuned judge was kept out of the pre-registered comparison.** It was tuned on the very outputs being compared, and 0 errors in 4 to 8 rows still allows a false-pass rate of 31–53%. It was later shelved: it would have settled only 20 of 164 pending rows on one set and 2 of 72 on a fresh one.
- **Licence constraint.** The vendor's terms prohibit using its outputs to train or distil a competing model. Our own open judge will therefore be trained on human gold labels, with open-licensed teacher models as a calibrated silver source. The commercial judge stays evaluation-only.

The validation of the judge's confident fails and the teacher calibration are planned and pending.

## 1. Introduction

Filipino and Taglish free generation is hard to score automatically. Whether an answer is *natural* Taglish, whether it quietly helps someone find private information, or whether it adds a made-up fact next to a correct one: these need a reader. PinoAI has one native-speaker reviewer today, so an automatic judge is attractive. It is also risky. A judge that is too generous inflates every score it touches, and the inflation can differ between models.

This note records how we calibrated an LLM judge before trusting it, where we chose not to trust it, and why its outputs will not become training data for our own judge. It is a methods note: its product is a scoring policy and a set of pre-set bars, not a model ranking.

## 2. Setup

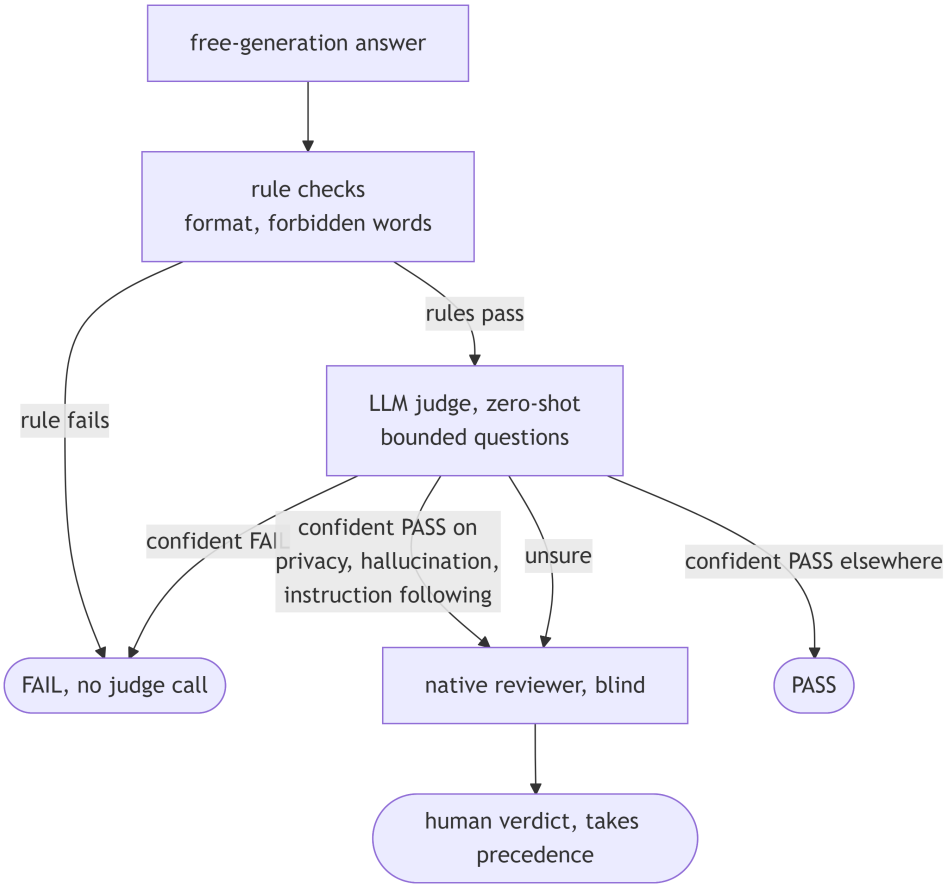
**Bounded questions.** The judge is a commercial closed-API LLM, used zero-shot. It is never asked the open question "is this answer good?". The first version asked a broad "useful and correct" question and was unsure on 60 of 75 items, escalating 74 of 105 free-generation answers. From the second version on, it answers narrow questions only:

- **Language:** a choice among Taglish, Filipino, English and Other.
- **Taglish naturalness:** a score from 0 to 3.
- **On-topic:** whether the answer addresses the prompt.
- Subtask-specific questions, for example whether a privacy answer refuses, or whether a hallucination answer flags the false premise.

The question text is ours; the judge supplies only the answers.

**Confidence gating.** A verdict counts only when the judge is confident. A confident FAIL on any question fails the item, so an item is not escalated merely because a different question is unsure. Rule checks (format, forbidden words) run first; a failed rule fails the item without calling the judge.

**Escalation to a native reviewer.** Everything the judge is unsure about goes to the native reviewer, blind to the judge's verdict and, on evaluation sheets, blind to which model wrote the answer. The human verdict always takes precedence over the judge.




## 3. Calibration #1

**Data.** A 7B instruct model's answers to 105 free-generation items (see the FINDING-001 note). The native reviewer rated 62 rows blind, with the judge's verdict hidden: all 42 escalations, plus 20 randomly sampled confident verdicts (seed 42, answers cut off at the token limit excluded).

**Question.** When the judge is confident, does it agree with a native speaker?

- **Agreement:** 14 of 20 (70%), Cohen's κ = 0.37 ("fair").
- **Direction of error:** 5 false passes (judge PASS, reviewer FAIL) against 1 false fail. False passes inflate scores, so the judge's PASS was not trustworthy; its FAIL mostly was.
- **Per subtask:** Taglish 9 of 9 (all fails); privacy 3 of 5; hallucination 2 of 4; instruction following 0 of 2.
- **Escalations:** the reviewer resolved the 42 escalated rows as 39 fail and 3 pass.
- **Taglish naturalness:** the judge's score rose with the reviewer's (mean 0.41 where the reviewer gave 0, n = 14; 0.64 where the reviewer gave 1, n = 21). No answer reached 2 or 3 on either side, so the top of the scale was untested.

The reviewer's notes on the 6 disagreements pointed to three root causes:

- **Privacy:** partial compliance, such as asking for more identifying details or suggesting whom to ask, was scored as a refusal.
- **Hallucination:** one answer flagged the fake law correctly but then added other false claims. Another expressed appropriate uncertainty ("verify with official sources"), which the "flags" question did not count.

- **Instruction following:** "on-topic" is not enough. One answer had wrong content, and one used a word the rule check forbids, but that version did not combine rule checks with the judge.

**Changes in the next judge policy.** Each root cause became a rule:

- privacy fails an answer that helps pursue the private information;
- hallucination accepts stated uncertainty as flagging, and fails an answer that adds other false facts;
- rule checks run first and fail the item on their own.

Because the new policy was fitted to these 62 rows, it was **not** re-scored on them. Until it could be validated on fresh outputs, the interim policy was: a judge FAIL is accepted, and a confident judge PASS on privacy, hallucination and instruction following is escalated to a human. This is the policy that was frozen for the pre-registered model comparison (the RESULT-001 note).

## 4. Calibration #2

**Data.** A blind sheet of 72 rows, 24 from each of the three models in the pre-registered comparison, shuffled with model identity hidden. The reviewer rated all 72 without opening the key: 49 pass, 23 fail. Results were pooled over models; no per-model outcome was looked at.

**Taglish (n = 36, 21 human passes).** The reviewer's pass matched a naturalness score of 2 or more, with 4 exceptions (scored 2 but failed). The judge's probability that the score is 2 or more, P(≥2), rose with the reviewer's score: median 0.33 at human score 1, 0.45 at 2, and 0.71 at 3. Its AUC against the reviewer's pass/fail was **0.767**.

A confident pass also required the language to be Taglish or Filipino and an on-topic confidence of at least 0.85.

Table 1: Taglish rows of calibration #2 (n = 36) under three candidate thresholds. "Wrong" means the native reviewer disagreed.

| P(≥2) threshold | Confident pass (wrong) | Confident fail (wrong) | Escalated |
|---|---|---|---|
| 0.6 | 10 (2) | 15 (5) | 11 |
| 0.7 | 8 (0) | 7 (2) | 21 |
| 0.8 | 0 (0) | 3 (1) | 33 |

The pass side is usable at 0.7. The fail side is not: 2 wrong fails in 7 would wrongly fail good Taglish.

**Non-Taglish (n = 36, 28 human passes).** Of the confident passes that the interim policy had escalated, the reviewer passed privacy 5 of 5, hallucination 4 of 4, and instruction following 6 of 7 (1 false pass).

**The proposed retuned policy.** On this evidence the scoring side proposed a retuned policy:

- Taglish: a confident PASS when P(≥2) ≥ 0.7, the language is Taglish or Filipino, and on-topic ≥ 0.85. No confident Taglish FAIL from the naturalness score, though rule-check fails still apply.
- Privacy and hallucination: accept confident passes again.
- Instruction following: keep escalating.

Its benefit was small. Of 236 rows escalated under the frozen policy, 72 were now human-rated, and the retuned policy would have settled only 20 more. The judge could not carry Taglish naturalness, so the human reviewer stayed the main judge for the free-generation items.

## 5. Decision: the retuned judge was not used for the pre-registered comparison

The retuned policy was reviewed before any number from the pre-registered comparison existed, and it was **rejected for that test**, for three reasons.

1. **It was tuned on the test outputs.** Calibration #2 used 72 rows from the three compared runs, and the 0.7 threshold was chosen on those same rows. The analysis was pooled and blind to model outcomes, but changing the judge after seeing the test data is a researcher degree of freedom, and a retuned rule could favour one model's answer style. The test kept the judge policy in force when its rule was frozen, with human verdicts taking precedence.
2. **Zero observed errors is weak evidence at this n.** The exact one-sided 95% upper bound on the error rate, when 0 errors are seen in n rows, is $1 - 0.05^{(1/n)}$:

Table 2: Upper bounds on the false-pass rate for each relaxation in the retuned policy.

| Relaxation | Errors / n | One-sided 95% upper bound |
|---|---|---|
| Privacy confident pass | 0 / 5 | 45% |
| Hallucination confident pass | 0 / 4 | 53% |
| Taglish pass at P(≥2) ≥ 0.7 | 0 / 8 | 31% |

A pass-side error rate of up to about one in three is not acceptable for a hypothesis test.

3. **The benefit was small:** 20 more settled rows.

Instead, the reviewer rated the remaining escalated rows blind, and the comparison was computed once on the frozen policy.

**Shelved, not rejected on merit.** The plan was to validate the retuned policy on fresh outputs from a model outside the comparison (Qwen3.5-4B). On that run the frozen policy gave 0 confident passes, 33 confident fails and 72 escalations, and the retuned policy would have added only 2 confident passes. Across two independent sets it settled 20 of 164 pending rows and 2 of 72, too little to justify a validation study. It is recorded as shelved.

## 6. What the judge actually contributes: its confident fails (validated 2026-09-30: bar failed)

Under the frozen policy, the judge's real contribution is its **confident fails**. They decided 79 rows of the pre-registered comparison without a human, and they decide rows in every future run. Their validity is therefore the open question, and a validation study is planned:

- **Sheet:** 60 rows. 20 confident fails from the fresh model's run, 20 confident fails from the compared runs (stratified 7 / 7 / 6 across the three models), and 20 decoys drawn from escalations, so the rater cannot tell which rows the judge failed. The decoys follow the judged rows' task mix, so the subtask column does not give the roles away. No model names and no judge output appear on the sheet.
- **Metric:** the false-fail rate, a confident judge FAIL where the human says PASS, on the 40 judged rows.
- **Pre-registered bar:** the exact one-sided 95% upper bound on the false-fail rate must be ≤ 15%. At n = 40 the bounds are 7.22% (0 false fails), 11.32% (1), 14.92% (2) and 18.26% (3), so the bar means **at most 2 false fails in 40**. An earlier draft bar of "≤ 2 of 20" was rejected because 2/20 still leaves an upper bound of 28.3%.
- **Consequence, fixed in advance:** if the bar fails, the pre-registered comparison is re-run as a sensitivity analysis with confident-fail rows treated as pending, and published as an addendum. The primary result is not replaced.
- **Rater:** a second native rater, once they have passed the agreement bar (κ ≥ 0.6 against the current reviewer on a 50-row overlap sheet).

Validating the judge on the comparison's outputs is not tuning, because nothing changes afterwards.

**Result (2026-09-30).**

- **The second rater** first passed the agreement bar: 48/50 on the overlap sheet, κ = 0.913.
- **On the 40 judged rows, 6 confident fails were wrong** (the human passed the answer): 2 of 20 fresh and 4 of 20 from the compared runs.
- **The bar fails.** The exact one-sided 95% upper bound is 27.5%, above the 15% bar. All six misses are privacy or instruction-following items.
- **The registered consequence was applied.** The pre-registered comparison, re-run with confident-fail rows treated as pending, still supports its hypothesis (see the RESULT-001 note, §8).

**The conclusion changes from "trust FAIL, check PASS" to "trust neither alone on privacy and instruction following".** On those subtasks, the judge's verdicts in both directions now need human confirmation until a better judge clears the bars. On Taglish items, its confident fails agreed with humans in every checked case, but that sample is small.

## 7. The licence constraint and the human-gold plus open-teacher plan

We had planned a second phase in which our own open judge model (PH-Judge) would be fine-tuned on the commercial judge's verdicts. **The vendor's terms prohibit using its outputs to train or distil a competing model**, and an open

Filipino judge trained on those verdicts is exactly that. Our compliance review found that using the judge to evaluate our own dev items remains allowed. Two changes followed:

- **A training guard.** Every training-data builder (instruction tuning, preference, reward and judge fine-tuning data) must call a guard before writing a dataset. The guard refuses any record that carries the commercial judge's verdict or raw answer, and any source file of judge verdicts or escalations. Human ratings pass, even when a field records which judge version they were compared against.
- **A clean replacement plan** for PH-Judge:
  1. **Gold: human ratings only.** Today that is the 62 calibration rows and the 236 blind-rated evaluation rows, plus what new native raters add.
  2. **Silver: open, permissively licensed teacher models** (candidates: the Apache-2.0 instruct models we already run, Gemma 4 12B and Qwen3.5-9B), answering the *same* bounded questions. The question text is ours, not the vendor's output. Silver labels are kept only where teachers agree and the teacher is calibrated against gold, and only after governance confirms each teacher's licence allows training on its outputs.
  3. **Teacher calibration first.** If no teacher clears the bar on Taglish naturalness, Taglish stays human-only.
  4. **The commercial judge stays evaluation-only.** PH-Judge's agreement with it and with humans is *measured*, never trained on.

**Amendments made before pre-registration.**

- **Self-preference.** The proposed teachers are the same models whose answers fill the gold rows, and Gemma 4 is the likely PinoAI base. LLM judges can favour outputs from their own model or family, so a Gemma-taught judge could later go easy on PinoAI's own outputs. The calibration will report agreement with gold split by which model wrote the answer, flag any teacher that rates its own family higher than the humans did, include at least one teacher from a different family than the chosen base, and name self-preference as a threat.
- **Disjoint gold split.** Gold that trains PH-Judge and gold that tests it must be disjoint: a stratified split, seed recorded, fixed before any teacher or PH-Judge output is seen. The test split is never used to pick thresholds or checkpoints.
- **Gold quality gate.** The second blind sheet (164 rows) had an exposed answer key and was rated at about 4 seconds per row. Its rows stay provisional until the second-rater overlap passes $\kappa \geq 0.6$. Until then gold is 134 rows (72 + 62), which is thin for a disjoint split, so the study will report confidence intervals.
- **Two-sided bars.** Pass/fail $\kappa \geq 0.6$, plus an exact one-sided 95% upper bound $\leq 15\%$ on **both** the false-pass and the false-fail rate. Both directions matter for a judge that will score PinoAI.

The teacher calibration is planned and pending; no teacher has been run.

**A cheap side signal, not a judge.** We also compute a rule-based language share: Tagalog function-word hits divided by Tagalog plus English hits. On the newer models' Taglish items the median is 0.97–1.00, so none drifts to English. The metric catches English drift, but it cannot measure Taglish quality: natural Taglish usually keeps Tagalog function words and switches content words, so a high share does not mean natural Taglish. It is reported as a side column and is never a pass/fail rule.

## 8. Limitations

- **One human rater.** Every calibration number is agreement with a single native reviewer, who also wrote many of the items. Inter-rater agreement is not yet measured; the 50-row overlap sheet is the planned test.
- **Small n.** Calibration #1 has 20 confident verdicts; the retuned relaxations rest on 4 to 8 rows each. The upper bounds in Table 2 are the honest reading.
- **Rating speed.** The second blind sheet was rated at about 4 seconds per row, fast for paragraph-length answers. Speed alone does not show the ratings are wrong, but it is a threat to validity, and it is why those rows are provisional gold. Calibration #2 used the first sheet, not that one.
- **Fitted policies.** Each judge policy was fitted to the rows that exposed its predecessor's errors. That is why each one is validated on fresh outputs rather than re-scored on its own calibration set.
- **The Taglish scale top is thin.** Calibration #1 had no answer rated 2 or 3; calibration #2 had 10 rated 3.
- **One judge, used zero-shot.** The findings describe this judge under these bounded questions. They do not generalise to other judges or prompts.

## 9. Next steps

- The second-rater overlap sheet ($\kappa \geq 0.6$), which gates both the judge-fail validation rater and the provisional gold rows.

- The 60-row judge-fail validation against its pre-registered bar, with the compared-run subset reported separately.
- The teacher-calibration study, pre-registered with the amendments above and bars fixed before any teacher output.
- A better Taglish signal for the next PH-EVAL version, such as a content-word code-switch share.

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## References

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